STAT 429: Introduction to Time Series
Week 11: Backshift, AR Models, and Partial Autocorrelation

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1 Backshift Notation

A convenient piece of notation used while working with AR and MA models is the Backshift notation. Let B denote the *backshift operator* defined by

$$By_t = y_{t-1}, B^2y_t = y_{t-2}, B^3y_t = y_{t-3}, \dots$$

and similarly

$$B\epsilon_t = \epsilon_{t-1}, B^2\epsilon_t = \epsilon_{t-2}, B^3\epsilon_t = \epsilon_{t-3}, \dots.$$

Also let I denote the identity operator: $Iy_t = y_t$. More generally, we can define polynomial functions of the Backshift operator by, for example,

$$(I + B + 3B^2)y_t = Iy_t + By_t + 3B^2y_t = y_t + y_{t-1} + 3y_{t-2}.$$

In general, for every polynomial $f(z)$, we can define $f(B)$. One can even extend this notation to negative powers of B which correspond to forward shifts. For example, $B^{-1}y_t = y_{t+1}, B^{-5}y_t = y_{t+5}$ and $(B^3 + 9B^{-2})y_t = y_{t-3} + 9y_{t+2}$ etc.

In this notation, the defining equation $y_t = \phi_0+\phi_1y_{t-1}+\phi_2y_{t-2}+\cdots+\phi_py_{t-p}+\epsilon_t$ for the $AR(p)$ model can be written as $\phi(B)y_t = \phi_0 + \epsilon_t$ for the polynomial $\phi(z) = 1 - \phi_1z - \phi_2z^2 - \cdots - \phi_pz^p$.

The defining equation $y_t = \epsilon_t + \theta_1\epsilon_{t-1}$ for the MA(1) model can be written as $y_t = \theta(B)\epsilon_t$ for the polynomial $\theta(z) = 1 + \theta_1z$.

The defining equation $y_t = \epsilon_t + \theta_1\epsilon_{t-1} + \cdots + \theta_q\epsilon_{t-q}$ for the MA(q) model becomes $y_t = \theta(B)\epsilon_t$ for the polynomial $\theta(z) = 1 + \theta_1z + \cdots + \theta_qz^q$.

2 Causal Stationary AR(1) Formula Using Backshift

A causal stationary AR(1) process, with $|\phi_1| < 1$, has the representation

$$y_t = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j}, \qquad \epsilon_t \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2). \tag{1}$$

Backshift notation provides a convenient way to obtain this formula and to extend it to higher-order AR models. First note that AR(1) difference equation in backshift notation is

$$\phi(B)y_t = \phi_0 + \epsilon_t \qquad \text{where } \phi(z) = 1 - \phi_1z.$$

Thus we can formally write

$$y_t = \frac{1}{\phi(B)}\left(\phi_0 + \epsilon_t\right).$$

When $|\phi_1 z| < 1$, the geometric series gives

$$\frac{1}{\phi(z)} = \frac{1}{1-\phi_1 z} = 1 + \phi_1 z + \phi_1^2 z^2 + \phi_1^3 z^3 + \ldots, \tag{2}$$

For $|\phi_1| < 1$, this suggests the convergent filter expansion

$$\begin{aligned} y_t &= \left(I + \phi_1 B + \phi_1^2 B^2 + \ldots\right)\left(\phi_0 + \epsilon_t\right) \\ &= \left(I + \phi_1 B + \phi_1^2 B^2 + \ldots\right)\phi_0 + \left(I + \phi_1 B + \phi_1^2 B^2 + \ldots\right)\epsilon_t \\ &= \left(1 + \phi_1 + \phi_1^2 + \ldots\right)\phi_0 + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j} = \frac{\phi_0}{1-\phi_1} + \sum_{j=0}^{\infty} \phi_1^j \epsilon_{t-j} \end{aligned}$$

which gives (1). This formal method is sometimes called Backshift Calculus and it works for higher order AR models as well.

3 AR(p) for $p \geq 1$

Consider the AR(p) difference equation. In backshift notation, this is

$$\phi(B) y_t = \phi_0 + \epsilon_t \qquad \text{where } \phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p.$$

We shall call $\phi(z) := 1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p$ the AR polynomial.

We "solve" the AR(p) by writing:

$$y_t = \frac{1}{\phi(B)}(\phi_0 + \epsilon_t) = \frac{1}{1 - \phi_1 B - \cdots - \phi_p B^p}(\phi_0 + \epsilon_t)$$

The next step is to make sense of $1/(1 - \phi_1 B - \cdots - \phi_p B^p)$. It is natural here to factorize the polynomial $1 - \phi_1 B - \cdots - \phi_p B^p$ into linear factors, and then use (2). So we write

$$\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p = (1 - a_1 z) \ldots (1 - a_p z). \tag{3}$$

so that

$$\phi(B) = (1 - a_1 B) \ldots (1 - a_p B).$$

The numbers $a_1, \ldots, a_p$ appearing in (3) are simply the reciprocals of the roots of $\phi(z)$ i.e., the roots of $\phi(z)$ are given by $1/a_1, \ldots, 1/a_p$. Note here that some of the a_j's can be complex because the polynomial $1 - \phi_1 z - \cdots - \phi_p z^p$ can have complex roots (even though all its coefficients are real).

We then get

$$y_t = \frac{1}{(1 - a_1 B) \ldots (1 - a_p B)}(\phi_0 + \epsilon_t) = \prod_{k=1}^{p} \frac{1}{1 - a_k B}(\phi_0 + \epsilon_t).$$

For each $1/(1 - a_j B)$, we use the formula (2) to get:

$$y_t = \prod_{k=1}^{p} \left(\sum_{j=0}^{\infty} a_k^j B^j \right) \left(\phi_0 + \epsilon_t\right). \tag{4}$$

Multiplying out the product $\prod_{k=1}^{p}\left(\sum_{j=0}^{\infty} a_k^j B^j\right)$, we get

$$\begin{aligned} y_t &= \prod_k \left(\sum_{j=0}^{\infty} a_k^j B^j\right)(\phi_0+\epsilon_t) \\ &= \left(\sum_{j_1=0}^{\infty} a_1^{j_1} B^{j_1}\right)\cdots\left(\sum_{j_p=0}^{\infty} a_p^{j_p} B^{j_p}\right)(\phi_0+\epsilon_t) \\ &= \left(\sum_{j_1=0}^{\infty}\cdots\sum_{j_p=0}^{\infty} a_1^{j_1}\ldots a_p^{j_p} B^{j_1+\cdots+j_p}\right)(\phi_0+\epsilon_t) \\ &= \phi_0\sum_{j_1=0}^{\infty}\cdots\sum_{j_p=0}^{\infty} a_1^{j_1}\ldots a_p^{j_p} + \sum_{j_1=0}^{\infty}\cdots\sum_{j_p=0}^{\infty} a_1^{j_1}\ldots a_p^{j_p}\epsilon_{t-j_1-\cdots-j_p}. \end{aligned} \tag{5}$$

The above expression involves powers of $a_1,\ldots,a_p$. For these powers to not explode, we need

$$|a_i|<1 \qquad \text{for each } i=1,\ldots,p. \tag{6}$$

Note that a_i can be complex so $|a_i|$ represents the modulus of a_j. When $|a_i|<1$, the powers $|a_i|^j$ decay rapidly in j which makes the infinite sums above well-defined.

The formula above writes y_t in terms of $\epsilon_t,\epsilon_{t-1},\ldots$. By collecting terms where $j_1+\cdots+j_p=j$ for each $j=0,1,\ldots$, we can write this solution as

$$y_t=\mu+\sum_{j=0}^{\infty}\psi_j\epsilon_{t-j} \tag{7}$$

where $\psi_0=1$ and $\mu=\phi_0/\phi(1)=\phi_0/(1-\phi_1-\cdots-\phi_p)$. It can be checked that this is a stationary time series (note it is also causal as y_t only depends on $\epsilon_t,\epsilon_{t-1},\ldots$).

The key condition here is (6). This is the analogue of the AR(1) condition $|\phi_1|<1$ for AR(p) when $p\geq 1$. Because the roots of the AR polynomial $\phi(z)=1-\phi_1 z-\cdots-\phi_p z^p$ are $1/a_1,\ldots,1/a_p$, the condition (6) is equivalent to assuming that all roots of the AR polynomial are strictly larger than 1 in modulus.

This analysis can be made rigorous to show the following:

1. When all roots of the AR polynomial are strictly larger than 1 in modulus, then there exists a unique causal stationary process $\{y_t\}$ which satisfies the AR(p) equation: $y_t=\phi_0+\phi_1 y_{t-1}+\cdots+\phi_p y_{t-p}+\epsilon_t$. This causal stationary solution is of the form (7) where the coefficients $\psi_0,\psi_1,\psi_2,\ldots$ are derived from (5).

2. When even one root of the AR polynomial has modulus ≤ 1, then there cannot exist a causal stationary solution to the AR equation.

3. `AutoReg` fits the model

   $$y_1,\ldots,y_p=\text{fixed at observed values} \quad\text{and}\quad y_t=\phi_0+\phi_1 y_{t-1}+\cdots+\phi_p y_{t-p}+\epsilon_t$$

   for $t=p+1,\ldots,n$, with independent Gaussian innovations $\epsilon_t\sim N(0,\sigma^2)$ independent of the initial observations. When the condition (6) is true, this model has similar behavior to (7) after the influence of the initial values has decayed. Fixing the initial observations does not itself produce an exactly stationary process.

For assessing causal-stationarity of the fitted model, the output summary of `AutoReg` gives the values of the moduli of the roots of the fitted AR polynomial.

4 Determining the AR Order

How to determine the correct order p for fitting the AR(p) model? For this, one commonly uses a quantity called the sample PACF (PACF stands for Partial AutoCorrelation Function).

The sample PACF is defined as follows: for $h \geq 1$,

$$\text{sample PACF}(h) = \text{estimate } \hat{\phi}_{h,h} \text{ of } \phi_{h,h} \text{ when AR}(h) \text{ is fit to the data}$$

If the sample PACF(h) becomes negligibly small after a particular p, this suggests that AR(p) is a good model for the data. Equivalently, one can inspect whether the last coefficient in a fitted AR(h) model is small relative to its uncertainty. This is a heuristic for identifying an appropriate order, rather than a definitive test.

It can happen that the sample PACF(h) for $h = 1, 2, \ldots, 11$ are all negligible but at $h = 12$, it is nonnegligible. In that case, we would be using AR(12). More specifically, we will use that value of p for which sample PACF(h) is negligible for all $h > p$.

Why should the quantity $\hat{\phi}_h$ (obtained by fitting AR(h) to the data) be called the Sample Partial Autocorrelation? The following discussion explains this through multiple linear regression.

5 Partial Correlation and Multiple Linear Regression

Ordinary correlation measures the association between two variables. Partial correlation measures the association remaining after removing the linear contribution of other variables from *both* variables. This distinction explains the word "partial" in PACF.

5.1 Population linear regression and partial correlation

Let X, Y and $Z = (Z_1, \ldots, Z_k)^T$ have finite second moments. Write $L(X \mid Z)$ for the population linear regression of X on Z:

$$L(X \mid Z) = a_0 + a^T Z, \qquad (a_0, a) = \operatorname*{argmin}_{b_0, b} \mathbb{E}(X - b_0 - b^T Z)^2.$$

If $\Sigma_Z = \mathrm{Cov}(Z, Z)$ is invertible, then

$$a = \Sigma_Z^{-1} \mathrm{Cov}(Z, X), \qquad a_0 = \mathbb{E}X - a^T \mathbb{E}Z.$$

The residual $X - L(X \mid Z)$ has mean zero and is uncorrelated with each Z_j. Define $L(Y \mid Z)$ in the same way and set

$$U = Y - L(Y \mid Z), \qquad V = X - L(X \mid Z).$$

Provided both residual variances are positive, the **partial correlation** of X and Y after adjusting for Z is

$$\rho_{Y,X \cdot Z} = \mathrm{Corr}(U, V). \tag{8}$$

Thus we remove what Z explains linearly from X and Y, and then correlate what remains.

Remark 5.1 (Partial correlation and conditional correlation)**.** *Partial correlation is a single population number, whereas the conditional correlation* $\mathrm{Corr}(X, Y \mid Z = z)$ *can vary with* z*. They need not agree. If* (X, Y, Z) *is jointly Gaussian with a nonsingular covariance matrix, the conditional means are linear and the conditional covariance matrix does not depend on* z*. In that case conditional correlation is constant and equals partial correlation.*

5.2 What does a coefficient in multiple regression measure?

In simple regression of Y on X, the slope is

$$b = \frac{\mathrm{Cov}(Y, X)}{\mathrm{Var}(X)} = \mathrm{Corr}(Y, X)\sqrt{\frac{\mathrm{Var}(Y)}{\mathrm{Var}(X)}}.$$

The analogous statement for multiple regression uses residuals. Consider the population regression of Y on $Z_1, \ldots, Z_k, X$:

$$L(Y \mid Z, X) = b_0 + b_Z^T Z + b_X X.$$

Its coefficient on X satisfies

$$\boxed{b_X = \frac{\mathrm{Cov}(U, V)}{\mathrm{Var}(V)} = \rho_{Y,X\cdot Z}\sqrt{\frac{\mathrm{Var}(U)}{\mathrm{Var}(V)}}.} \tag{9}$$

This is the population version of the Frisch–Waugh–Lovell identity. It says that the coefficient on X is obtained by regressing the residualized response U on the residualized predictor V.

To see this directly, let $e = Y - b_0 - b_Z^T Z - b_X X$ be the full regression residual. It is orthogonal to $1, Z_1, \ldots, Z_k, X$, and hence to V. Removing the linear projection on $1, Z$ from the regression equation gives

$$U = b_X V + e.$$

Taking covariance with V yields $\mathrm{Cov}(U, V) = b_X \mathrm{Var}(V)$, proving (9).

A coefficient is not generally a partial correlation. The additional residual standard-deviation ratio in (9) is essential. The special symmetry of stationary time series will make that ratio equal to one.

6 The Population PACF and the Last AR Coefficient

Assume $\{y_t\}$ is a real-valued weakly stationary process with finite second moments and nonsingular covariance matrices for the blocks under consideration. No AR model is needed for the definition below. For $h \geq 2$, let

$$Z = (y_{t-1}, \ldots, y_{t-h+1})^T, \qquad U = y_t - L(y_t \mid Z), \qquad V = y_{t-h} - L(y_{t-h} \mid Z).$$

The **partial autocorrelation at lag** h is

$$\alpha(h) = \mathrm{Corr}(U, V). \tag{10}$$

For $h = 1$, there are no intervening variables, so $\alpha(1) = \mathrm{Corr}(y_t, y_{t-1}) = \rho(1)$. Weak stationarity ensures that these definitions do not depend on t.

The question answered by $\alpha(h)$ is: after removing the linear information in the intermediate observations, how much association remains between y_t and y_{t-h}?

6.1 Why are the two residual variances equal?

Write $\gamma(j) = \mathrm{Cov}(y_t, y_{t-j})$ and $\Gamma = \mathrm{Cov}(Z, Z)$. Let

$$c = \mathrm{Cov}(Z, y_t) = (\gamma(1), \ldots, \gamma(h-1))^T.$$

The covariance vector for the other endpoint is the reverse vector:

$$d = \mathrm{Cov}(Z, y_{t-h}) = (\gamma(h-1), \ldots, \gamma(1))^T = Jc,$$

where J reverses the order of a vector. Stationarity gives $\Gamma_{ij} = \gamma(i-j)$, so $J\Gamma J = \Gamma$ and $J\Gamma^{-1}J = \Gamma^{-1}$. The usual regression residual-variance formula therefore gives

$$\begin{aligned}\mathrm{Var}(U) &= \gamma(0) - c^T\Gamma^{-1}c,\\ \mathrm{Var}(V) &= \gamma(0) - d^T\Gamma^{-1}d = \gamma(0) - c^TJ\Gamma^{-1}Jc = \mathrm{Var}(U).\end{aligned}$$

For $h = 1$, the equality follows simply from $\mathrm{Var}(y_t) = \mathrm{Var}(y_{t-1})$.

This is a symmetry of the covariance matrix under time reversal; weak stationarity alone is sufficient. A stationary Gaussian process also has full distributional time-reversal symmetry, but Gaussianity is not needed for this residual-variance argument.

6.2 The last coefficient is exactly the PACF

For each $h \geq 1$, consider the population regression

$$y_t = a_h + \phi_{h,1}y_{t-1} + \cdots + \phi_{h,h}y_{t-h} + e_{t,h}. \tag{11}$$

The first subscript records how many lags are included: the coefficients can change when h changes. Apply (9) with response y_t, last predictor y_{t-h}, and other predictors Z. Because the two residual variances agree,

$$\boxed{\phi_{h,h} = \frac{\mathrm{Cov}(U,V)}{\mathrm{Var}(V)} = \mathrm{Corr}(U,V) = \alpha(h).} \tag{12}$$

This is the population justification for estimating the PACF using the last coefficient in an AR(h) regression.

6.3 Why does the PACF cut off after the true AR order?

Suppose now that the process is a causal stationary AR(p):

$$y_t = \phi_0 + \phi_1y_{t-1} + \cdots + \phi_py_{t-p} + \epsilon_t,$$

where the innovation ϵ_t has mean zero and is independent of past observations. If $h > p$, the intervening variables Z already contain $y_{t-1}, \ldots, y_{t-p}$. Therefore

$$L(y_t \mid Z) = \phi_0 + \phi_1y_{t-1} + \cdots + \phi_py_{t-p}, \qquad U = \epsilon_t.$$

The other residual V is a linear combination of past observations, so $\mathrm{Cov}(\epsilon_t, V) = 0$. Consequently,

$$\boxed{\alpha(h) = 0 \quad \text{for every } h > p.}$$

Equivalently, in the larger population regression (11), the extra coefficients beyond lag p are zero. At lag p itself, $\alpha(p) = \phi_p$, which is nonzero if p is the true order. There need not be a nonzero PACF at every lag below p.

For a stationary AR(1), for example,

$$\rho(h) = \phi_1^h \quad (h \geq 0), \qquad \alpha(1) = \phi_1, \qquad \alpha(h) = 0 \quad (h \geq 2).$$

Observations two or more steps apart can remain correlated because information passes through intermediate observations. Partial correlation removes that linear contribution.

7 Sample PACF and the Last Fitted AR Coefficient

For observations $y_1, \ldots, y_n$, fit by ordinary least squares

$$y_t = a_h + \phi_{h,1} y_{t-1} + \cdots + \phi_{h,h} y_{t-h} + e_t, \qquad t = h+1, \ldots, n.$$

The **OLS sample PACF** is defined by

$$\boxed{\widehat{\alpha}_{\mathrm{OLS}}(h) = \widehat{\phi}_{h,h}.}$$

When the order h is understood, the last coefficient is often written simply as $\widehat{\phi}_h$. Each lag h uses its own regression with h predictors. One does not fit a single large AR model and interpret all its coefficients as PACF values.

The justification is now explicit: the population target of this last coefficient is $\phi_{h,h}$, and (12) proves that this equals the population PACF $\alpha(h)$.

7.1 The exact sample multiple-regression identity

Use the same rows $t = h+1, \ldots, n$ for all regressions in this paragraph. Regress y_t on an intercept and the intermediate lags to obtain residuals $\widehat{U}_t$. Separately regress y_{t-h} on an intercept and those same intermediate lags to obtain residuals $\widehat{V}_t$. The sample Frisch–Waugh–Lovell identity gives

$$\begin{aligned} \widehat{\phi}_{h,h} &= \frac{\sum_t \widehat{U}_t \widehat{V}_t}{\sum_t \widehat{V}_t^2} \\ &= \mathrm{Corr}_{\mathrm{sample}}(\widehat{U}, \widehat{V}) \sqrt{\frac{\sum_t \widehat{U}_t^2}{\sum_t \widehat{V}_t^2}}. \end{aligned}$$

Thus directly computing the correlation between these two sample residual vectors does not generally give exactly the OLS coefficient. Their sample variances need not be identical: stationarity is a population property, and a finite observed block need not have exact reversal symmetry. Under the usual consistency conditions for these sample moments, the ratio approaches one, matching the population identity.

Remark 7.1 (Which sample PACF does software compute?)**.** *Several estimators target the same population PACF. OLS uses the last coefficient in the lag regression; Yule–Walker methods use estimated autocovariances and a Toeplitz covariance matrix. Their finite-sample values can differ. In* `statsmodels`*, specifying* `pacf(y, nlags=H, method='ols')` *selects the OLS version discussed here. See the official documentation in the reference below.*

7.2 Using the plot to choose an order

Plot $\widehat{\alpha}(h)$ over a suitable range of lags. A pattern in which values beyond lag p fluctuate near zero supports an AR(p) model. A small value at one lag does not establish a cutoff: a larger value can appear at a later lag, for example at a seasonal lag such as 12. The population cutoff is exact for a causal stationary AR(p); its sample counterpart is a noisy diagnostic, to be assessed together with predictive performance and checks of the fitted residuals.

References

[1] statsmodels. Documentation for pacf and pacf_ols.
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